



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC3170
Course title (English)	Database System
Course title (Chinese)	数据库系统
Units	3
Language of Instruction	English
Description (English)	This is an introductory course in database systems. Topics discussed include database schemas and architectures, relational database, SQL, E-R-model, database design and normalization, indexing and hashing, transaction management, analytic processing, data warehouse and data mining.
Description (Chinese)	此课程为数据库系统的入门。讨论的主题包括系统结构, 数据库模型, 关系数据库, E-R模型, SQL语言, 关系数据库规范化, 索引和文件管理, 交易程序, 分析处理, 数据仓库和资料挖掘。

2. Prerequisites / Co-requisites

A. Prerequisites

CSC3100 Data Structures

B. Co-requisites

3. Learning Outcomes

To apply SQL to relational data definition and manipulation (K, S)
To develop an understanding of functional dependency and normalization(K, S)
To perform effective relational database design (S, V)
To perform conceptual database design using the E-R Model (K,S)
To transform E-R models into relational database schemas(K, S)
To explain the key functions of database transaction management(K)



To explain the difference between transaction processing and analytic processing (K, V)
To develop an understanding of data warehouse, business intelligence, and data mining(K,S, V)

4. Course syllabus

Database Schemas and Architectures, SQL, Relational Model and Functional Dependency, The Entity-Relationship Model, Database Design with and Normal Forms, Transformation of Data Models into Design, Indexing and Hashing, Transaction Management, Analytic Processing, Data Warehouse and Data Mining.

5. Assessment Scheme

Component/ method	% weight
Assignments	15
Midterm Test	20
Project	15
Final Examination	50

6. Grade descriptor

General

Grade	Description
A	Demonstrate an excellent level of understanding of the materials covered in the course in the final examination, midterm test, project and assignments.
A-	Demonstrate a high level of understanding of the materials covered in the course in the final examination, midterm test, project and assignments.
B	Demonstrate a sound understanding of the materials covered in the course in the final examination, midterm test, project and assignments.
C	Demonstrate a reasonable level of understanding of the materials covered in the course in the final examination, midterm test, project and assignments.
D	Demonstrate an acceptable level of understanding of the materials covered in the course in the final examination, midterm test, project and assignments.
F	Demonstrate minimal understanding of the materials covered in the course in the final examination, midterm test, project and assignments.



7. Feedback for evaluation

Informal interaction between the students and the instructor including face to face communication and email discussion

CTE at the end of the term

Student and advisor meetings

Program review

8. Reading

Required Reading

1.Database System Concepts, 9780078022159, A. Silberschatz, H. Korth, and S. Sudarshan, McGraw-Hill, 7TH EDITION, 2020, United States

9. Course components

Activity	Hours/week
Lectures	3
Tutorials	1

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Database Schemas and Architectures
002	The Entity-Relationship Model
003	The Relational Model
004	Functional and Multivalued Dependencies
005	Database Design and Normalization
006	SQL Fundamentals
007	Advanced SQL
008	Data Storage Structures
009	Indexing and Hashing
010	Transaction Management
011	Analytic Processing
012	Data Warehouse and Data Mining



013	Project Presentation
014	Revision

11. Any other information

12. Version date

Version number	6
As of (date)	2024/11/20